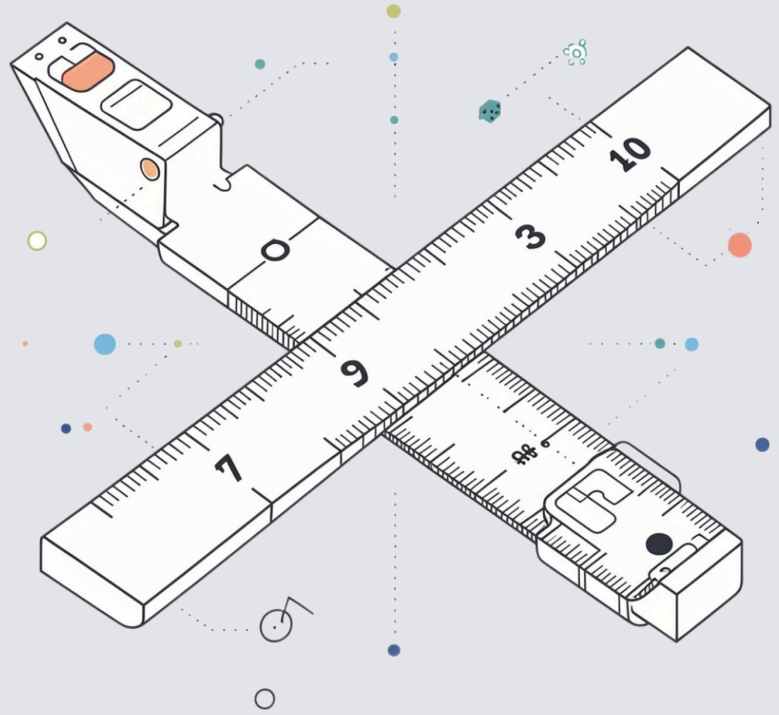
# SI Units & Unit Conversion

Class 11 Physics · 30-Minute Lecture



HOOK QUESTION

# Why can't we simply say the speed of a car is 20?

*What's missing?*

Think about it — 20 what? Miles? Kilometers? Meters per second?

A number alone is meaningless without a unit.

# What is a Measurement?

### Number

The magnitude — how much

### Unit

The standard — of what

### Together

5 meters · 10 seconds · 2 kilograms

# The 7 SI Base Units

| Quantity | Unit | Symbol |
| --- | --- | --- |
| Length | Meter | m |
| Mass | Kilogram | kg |
| Time | Second | s |
| Temperature | Kelvin | K |
| Electric Current | Ampere | A |
| Luminous Intensity | Candela | cd |
| Amount of Substance | Mole | mol |

# Derived Units

**Definition:** Units formed by combining SI base units through multiplication or division.

### Force – Newton (N)

$kg \cdot m/s^2$

### Energy – Joule (J)

$kg \cdot m^2/s^2$

### Power – Watt (W)

$kg \cdot m^2/s^3$

# Why Convert Units?

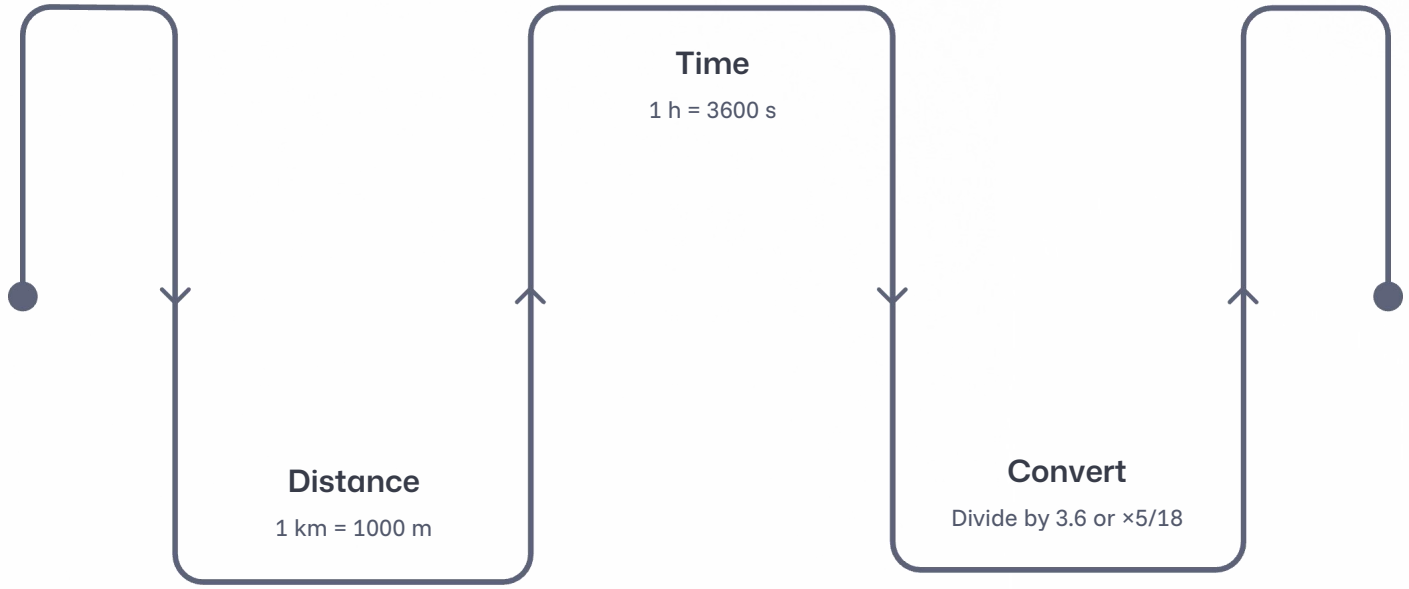
## The Problem

Real-world data arrives in mixed, inconsistent units

## The Requirement

Physics equations demand complete unit consistency on both sides

# Converting km/h → m/s




DERIVE ON BOARD

## Key Relation

**1 km/h = 5/18 m/s ≈ 0.278 m/s**

## Notes

Follow the derivation on the board step by step.

DERIVE ON BOARD

# Converting m/s → km/h

*Reciprocal of the previous conversion*

## Key Relation

**1 m/s = 18/5 km/h = 3.6 km/h**

Multiply by 3.6 to convert from m/s to km/h.

## Quick Reference

→ **km/h to m/s:** × 5/18

→ **m/s to km/h:** × 18/5

→ **m/s to km/h:** × 3.6

## My Notes

# Practice: Speed Conversions

1

**72 km/h = ? m/s**

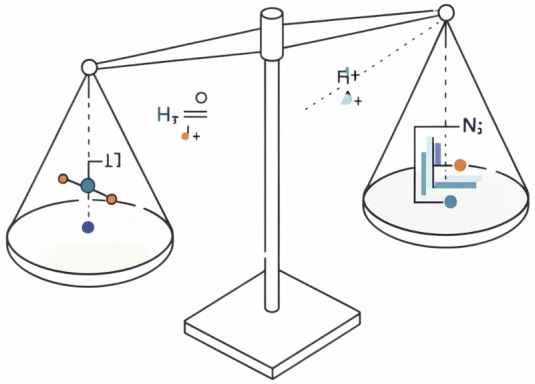
72 × (5/18) = **20 m/s**

2

**10 m/s = ? km/h**

10 × (18/5) = **36 km/h**

*Space for your working below ↓*



# Dimensional Analysis

**Rule:** Both sides of any equation must carry the same dimensions.

Dimensions don't lie — use them to verify every formula.

## Check F = ma

$[\text{kg} \cdot \text{m/s}^2] = [\text{kg}] \times [\text{m/s}^2]$ ✓

## Notes

# Exit Question

> *Why can't we simply say the speed of a car is 20?*

Write your answer in the space below — in one complete sentence, explain what makes a measurement meaningful.